

# Advanced NLP Exercise 1: Open Questions & Qualitative Analysis

Yoel Marcu

May 17, 2026

## Part 1: Open Questions (Section 1)

### Question 1: QA Datasets for Intrinsic Concepts

Question Answering is often used as a framework to test core linguistic structures (intrinsic concepts) rather than just factual knowledge. Here are three examples:

#### 1. QA-SRL (Semantic Role Labeling via Question Answering)

- **Explanation:** This dataset annotates predicate-argument structures by asking questions like "Who did what to whom?" based on verbs in a sentence. It measures an intrinsic property because it evaluates the model's grasp of core semantic roles and sentence structure, independent of external world knowledge.

#### 2. QAMR (Question-Answer Meaning Representation)

- **Explanation:** Expanding on QA-SRL, QAMR uses QA pairs to capture the relationships between *all* words in a sentence, including nouns and adjectives (e.g., "What kind of car?"). It intrinsically measures whether a model understands the complete, granular semantic dependency graph of a sentence.

#### 3. QA-Coref (Coreference Resolution via QA) / CorefQA

- **Explanation:** This method frames coreference resolution as a reading comprehension task by asking questions to link pronouns or noun phrases to their original entities (e.g., "Who does 'she' refer to?"). It measures the intrinsic linguistic ability to track entities and maintain grammatical coherence across a longer text.

### Question 2: Inference-Time Scaling

#### (a) Methods for Inference-Time Scaling:

##### 1. Self-Consistency (Majority Voting)

- **Description:** Instead of greedy decoding (taking the single most likely path), the model generates multiple different reasoning paths (Chain-of-Thought) for the same prompt and selects the final answer that appears most frequently among the outputs.
- **Advantages:** Significantly improves accuracy on reasoning and math tasks by smoothing out random errors in individual reasoning chains.
- **Bottlenecks:** Heavily consumes **compute (FLOPs)**, as the model must generate full responses  $N$  times.
- **Parallelizable? Yes.** All  $N$  paths are completely independent and can be generated simultaneously across multiple GPUs or in a single large batch.

##### 2. Tree of Thoughts (ToT)

- **Description:** The model maintains a "tree" of intermediate reasoning steps. It uses search algorithms (like Breadth-First or Depth-First Search) and a verifier/heuristic to evaluate branches, expanding promising steps and backtracking from dead ends.
- **Advantages:** Allows the model to perform deliberate planning and recover from mistakes, making it highly effective for complex, multi-step logic puzzles.
- **Bottlenecks:** Heavily consumes **memory (VRAM)** to store the tree states (KV caches of different branches) and **compute** to continuously run the verifier model on intermediate steps.
- **Parallelizable? Partially.** Branches at the same depth level can be evaluated in parallel, but the overall process is sequential due to the nature of the search algorithm.

### 3. Best-of-N (BoN) / Reranking

- **Description:** The model generates  $N$  complete, distinct solutions. A separate "Reward Model" or Verifier then scores all  $N$  solutions, and the one with the highest score is selected.
- **Advantages:** Excellent for tasks where verifying an answer is much easier than generating it (e.g., coding, where a unit test can act as the verifier).
- **Bottlenecks:** Consumes significant **compute** for both generation and the subsequent evaluation by the reward model.
- **Parallelizable? Yes.** The generation of the  $N$  candidate solutions, and the scoring of those solutions by the reward model, can be heavily parallelized.

#### (b) Scenario: Single GPU with Large Memory for Complex Scientific Reasoning

I would choose **Tree of Thoughts (ToT)** (or a highly batched implementation of Self-Consistency).

**Why?** Scientific reasoning requires complex, multi-step logic where a single early mistake ruins the entire derivation. ToT allows the model to backtrack rather than waste compute finishing a fundamentally flawed reasoning chain. Furthermore, the specific hardware constraint is a *single GPU* (limiting sheer parallel compute speed) but with *large memory capacity*. ToT is naturally memory-hungry because it has to hold the KV cache for multiple branching states simultaneously. A large-memory GPU is perfectly suited to absorb this bottleneck, allowing deep search trees without running out of VRAM.

## Part 2: Qualitative Analysis (Section 2.1, part 2)

**Did the configuration that achieved the best validation accuracy also achieve the best test accuracy?**

No. The configuration that achieved the highest validation accuracy (Run 2: 3 epochs, lr=3e-5) achieved a test accuracy of 83.25%. Surprisingly, the model with the lowest validation accuracy (Run 3: 5 epochs, lr=3e-5) actually achieved the highest test accuracy at 84.00%.

### **Analysis of Best (3-Epoch) vs. Worst (5-Epoch) Configurations:**

The configuration with the highest validation accuracy (86.52%) trained for 3 epochs, while the worst performing configuration (85.04%) trained for 5 epochs. Comparing validation examples where the 3-epoch model succeeded but the 5-epoch model failed reveals that the lower-performing model suffered from **overfitting**, resulting in two distinct patterns of failure:

### **1. The "Lexical Overlap" Trap (Ignoring Semantic Meaning)**

The 5-epoch model frequently misclassified sentences as paraphrases (Label: 1) when they were actually different (Label: 0) simply because they shared a high volume of identical vocabulary.

- **Example:**

- *Sentence 1:* Vivace was founded in 1999 and has raised over \$118 million in three rounds of venture financing.
- *Sentence 2:* During difficult times for technology venture capital, Vivace raised over \$118 million in three rounds of venture financing.

- **Analysis:** The overfit model memorized that heavy keyword matching (e.g., "Vivace", "\$118 million", "three rounds of venture financing") correlates strongly with paraphrases in the training set. It became "lazy" and relied on this word-overlap heuristic, failing to notice the massive semantic difference (one discusses the founding year, the other discusses difficult times for capital). The 3-epoch model retained enough generalization to read the actual semantic meaning of the differing clauses.

### **2. Structural Rigidity**

Conversely, the 5-epoch model failed to recognize actual paraphrases (Label: 1) if the sentence structure was significantly inverted or relied on heavy synonym use.

- **Example:**

- *Sentence 1:* Per-user pricing is \$29 for Workplace Messaging, \$89 for Team Collaboration and \$35 for Collaborative Learning.
- *Sentence 2:* Workplace Messaging is \$29, Workplace Team Collaboration is \$89, and Collaborative Learning is \$35.

- **Analysis:** Here, the exact same information is presented, but the grammar is inverted (Price → Product vs. Product → Price). Because the 5-epoch model memorized rigid syntactic structures from the training data, this structural inversion caused it to predict that they were not paraphrases (Label: 0). The 3-epoch model successfully recognized that despite the grammatical flip, the semantic relationships remained identical.